Beyond Version Control

GitHub Actions for Power BI Development

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Speaker info

- Budapest, Hungary
- BI Architect @ VISUAL LABS

✋ Power BI & Git - Where Are You on the Journey?

1. **I develop Power BI reports** — not just view them.
2. **I've struggled with version control** (Project_Final_v7_REAL_FINAL.pbix).
3. **I already work with .PBIP** projects instead of only .PBIX files.
4. **I've opened or edited a TMDL** file and understood (most of) what I saw.
5. **I've tested** my model against **the Best Practice Analyzer** – or I know it.
6. **I use Git** or another **version control** system **for my Power BI** projects.
7. **I've created or reviewed a Pull Request** (PR) before merging code.
8. **I've seen** or written a **GitHub Actions** or Azure DevOps YAML workflow.

Scan to refresh your Git, CI/CD, and Power BI knowledge

🎯 Main goals of this session to show how to:

1) 🌱 Use PBIP & Git to bring structure to Power BI development.

2) 🌿 Run BPA with GitHub Actions for fast, actionable feedback.

3) 🌳 Centralize and reuse workflows for consistent CI/CD.

Agenda

1. Dark Age – Before version control
2. Version control & co-op development
3. Automated quality checks with BPA & GitHub Actions
4. **Scalable CI/CD workflows**

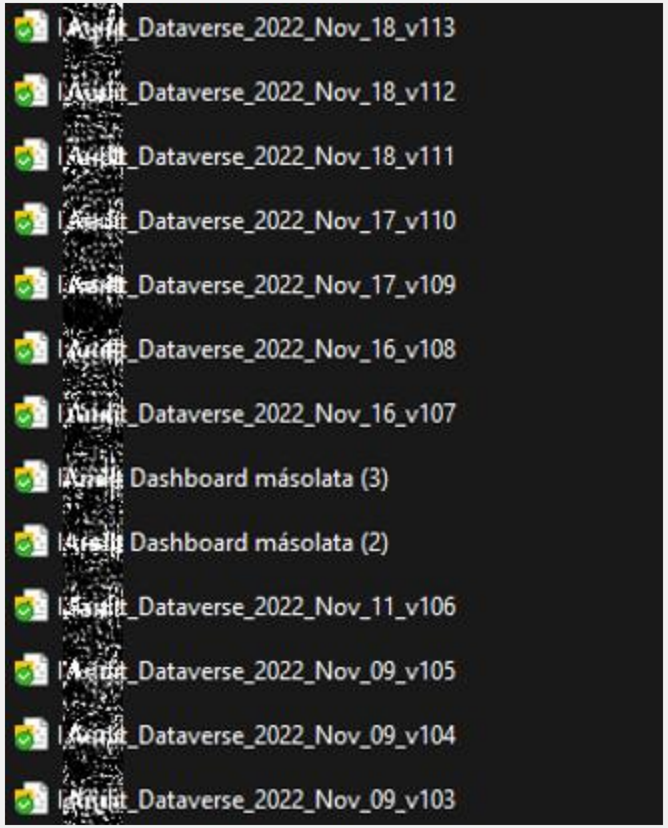
Dark Age

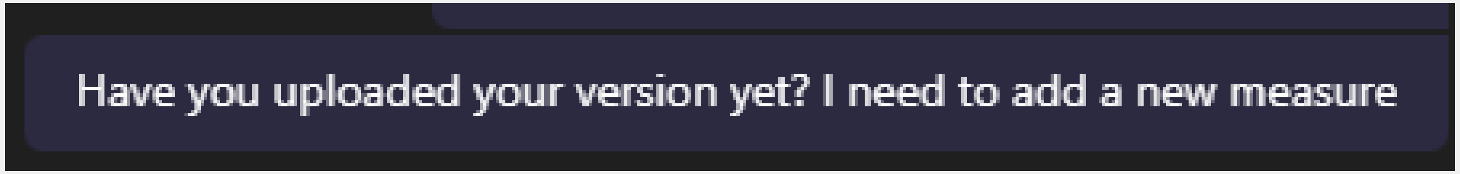
Before version control

The Dark Age of Power BI Version Control

Developing on your own

Developing in teams



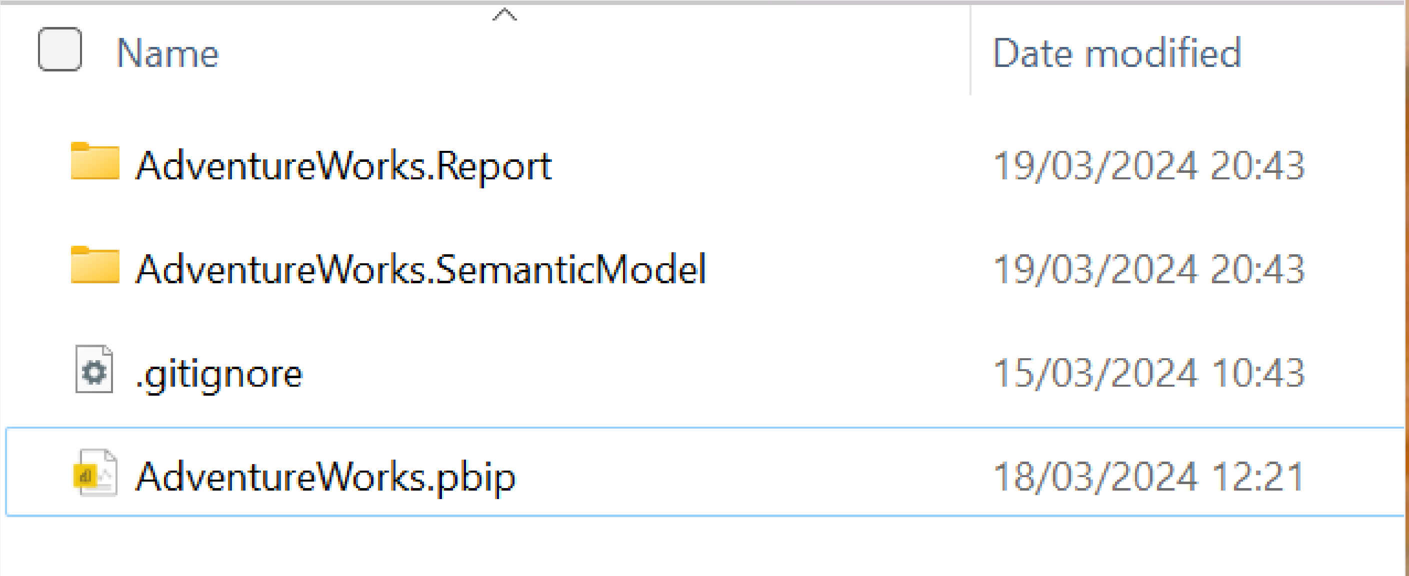


Version control

The start of a new era

Power BI Project Files

- From .pbix to .pbib



- Option to switch to TMDL format

CONFERENCE DEMO

- Demo Report.Report
  - .pbi
  - StaticResources
  - .platform
  - definition.pbir
  - report.json
- Demo Report.SemanticModel
  - .pbi
  - definition
    - cultures
    - tables
      - _Measures.tmdl
      - Campaign Returns.tmdl
      - CS_NewTickets.tmdl
      - Historical_Trend.tmdl
      - Marketing_Funnel.tmdl
    - database.tmdl
    - model.tmdl
  - .platform
  - definition.pbism
  - diagramLayout.json
- .gitignore
- Demo Report.pbip

OUTLINE

TIMELINE

- Demo Report.SemanticModel > definition > tables > Historical_Trend.tmdl

```
table Historical_Trend
	column Opened
		lineageTag: 27c31e28-e481-4ae6-8e49-560555d1ad63
		summarizeBy: sum
		sourceColumn: Opened

		changedProperty = DataType

		annotation SummarizationSetBy = Automatic

	column 'Converted Users'
		dataType: int64
		formatString: 0
		lineageTag: d875e020-740b-4432-a433-0cc4a86eeb3b
		summarizeBy: sum
		sourceColumn: Converted Users

		annotation SummarizationSetBy = Automatic

	column 'Converted value'
		dataType: int64
		formatString: 0
		lineageTag: adfbde04-734f-44e1-ba75-c1beea87094a
		summarizeBy: sum
		sourceColumn: Converted value

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	column 'Avg Conversion Value'
		dataType: int64
		formatString: 0
		lineageTag: de3bf1d3-f77f-40fd-98a8-e9f17845a8a6
		summarizeBy: sum
		sourceColumn: Avg Conversion Value

		annotation SummarizationSetBy = Automatic

	partition Historical_Trend = m
		mode: import
		source =
				let
					Source = Table.FromRows(Json.Document(Binary.Decompress(Binary.FromText("PVPJscUwCOslZw5m8UItmdd/Gx8J/xzCjI0MkiDv++gjz6jPENRwtC0x1/OT9zFc6qyYp0L4RI
					#"Changed Type" = Table.TransformColumnTypes(Source,{{"Converted Users", Int64.Type}, {"Converted value", Int64.Type}}),
					#"Added Custom" = Table.AddColumn(#"Changed Type", "Avg Conversion Value", each [Converted value] / [Converted Users]),
					#"Changed Type1" = Table.TransformColumnTypes(#"Added Custom",{{"Avg Conversion Value", Int64.Type}})
				in
					#"Changed Type1"

	annotation PBI_ResultType = Table

	annotation PBI_NavigationStepName = Navigation
```

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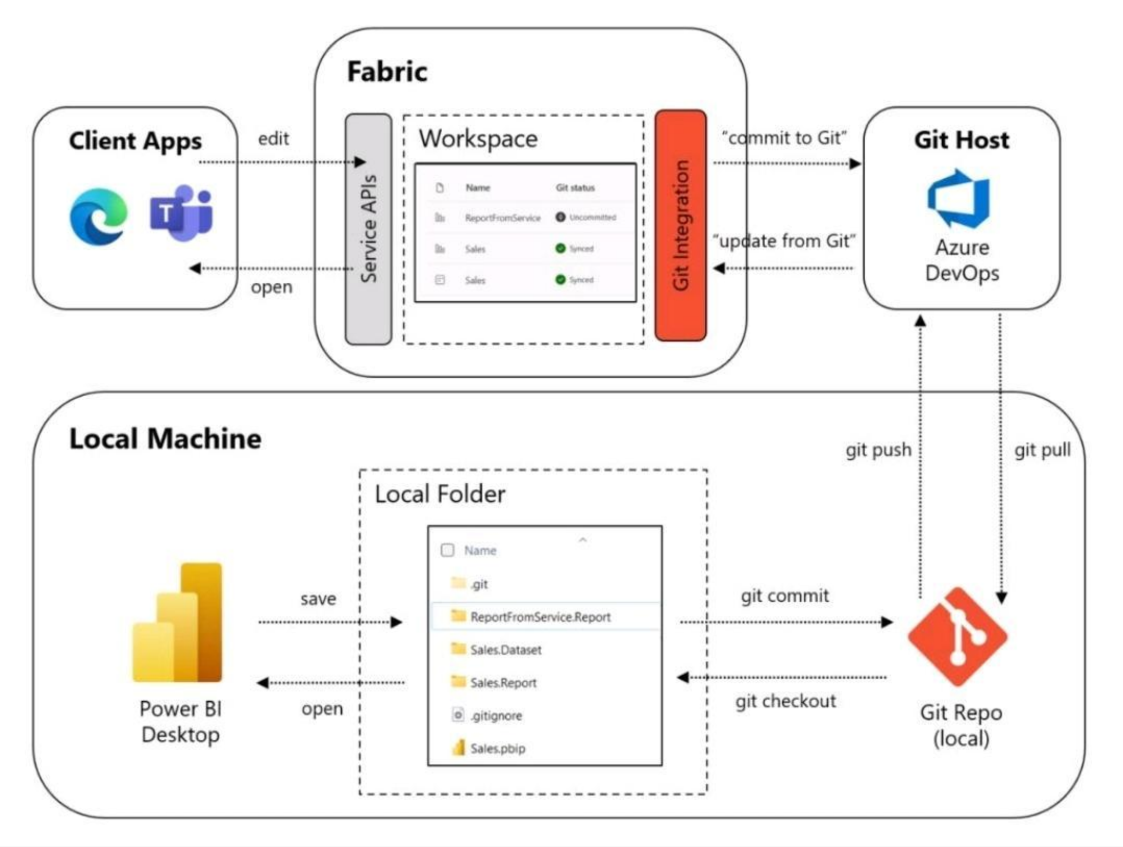
Git Basics



Fabric
Client Apps
edit
open
Service APIs
Workspace
Name
Git status
ReportFromService
Uncommitted
Sales
Synced
Sales
Synced
Git Integration
"commit to Git"
"update from Git"
Git Host
Azure
DevOps
git push
git pull
Local Machine
Local Folder
Name
.git
ReportFromService.Report
Sales.Dataset
Sales.Report
.gitignore
Sales.pbip
save
open
Power BI
Desktop
git commit
git checkout
Git Repo
(local)

Git basics

1) **Repo –** stores your project related files and track all changes made to files

2) **Commit –** snapshot of the current status of your project

3) **Branch –** separate workspace wher you candevelop and try new things w/o messing up the main project

4) **Merge –** integrate two different versions into one

5) **Pull Request (PR) –** formal way to ask for approval an merge changes of 2 branches

Your next challenge: How do you...

1. Ensure consistency across multiple Power BI projects?
2. Stop suboptimal practices from leaking into production?
3. Align developers with different habits and skill levels?
4. Automate security and quality checks reliably?

Beyond version control: PBI meets CI/CD

- Apply security measure on your main branch(es)
- Run automated tests
- *Automate security checks and approvals*

- Use familiar 3rd party tools like: Tabular Editor and run:
  - Best Practice Analyzer
  - Format DAX
  - Any macros

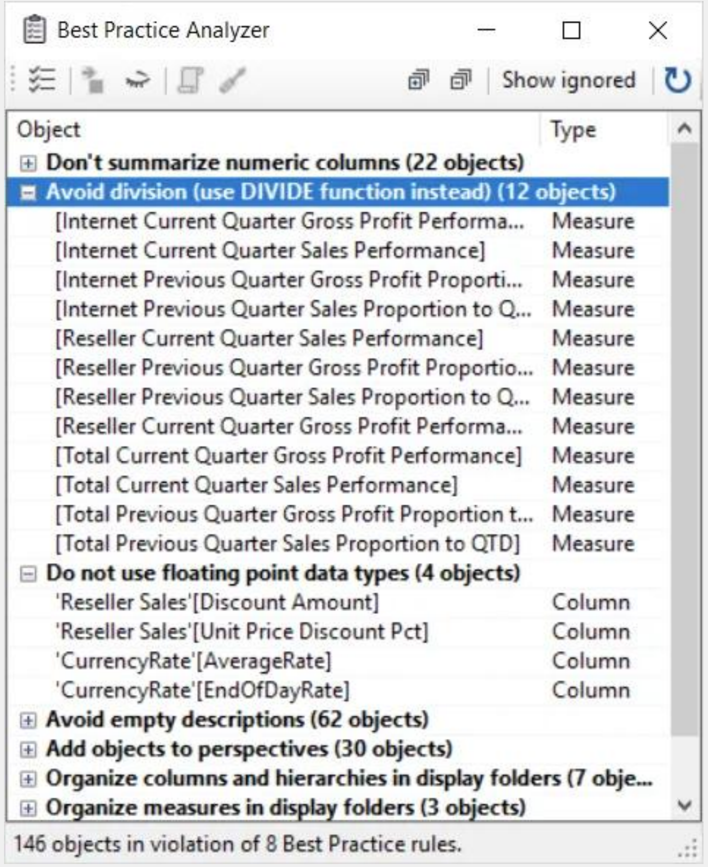
Beyond version control: PBI meets CI/CD

BPA Severity Levels
(higher severity = greater impact)

Severity 1: ℹ️ **Information:** There are Row-Level Security (RLS) roles with no members in your model.

Severity 2: ⚠️ **Warning:** There are two measures with different names but defined by the same DAX expression - this should be avoided to reduce redundancy.

Severity 3: 🛑 **Error:** There are numeric columns in your model with default summarization.



🌿 BPA & GitHub Actions

Automated quality checks

Basic demo scenario

1) Someone develops a new measures and wants to push it to the *main* branch

2) The developer creates a Pull Request (PR)

3) The PR triggers a GitHub Actions workflow which will run the BPA

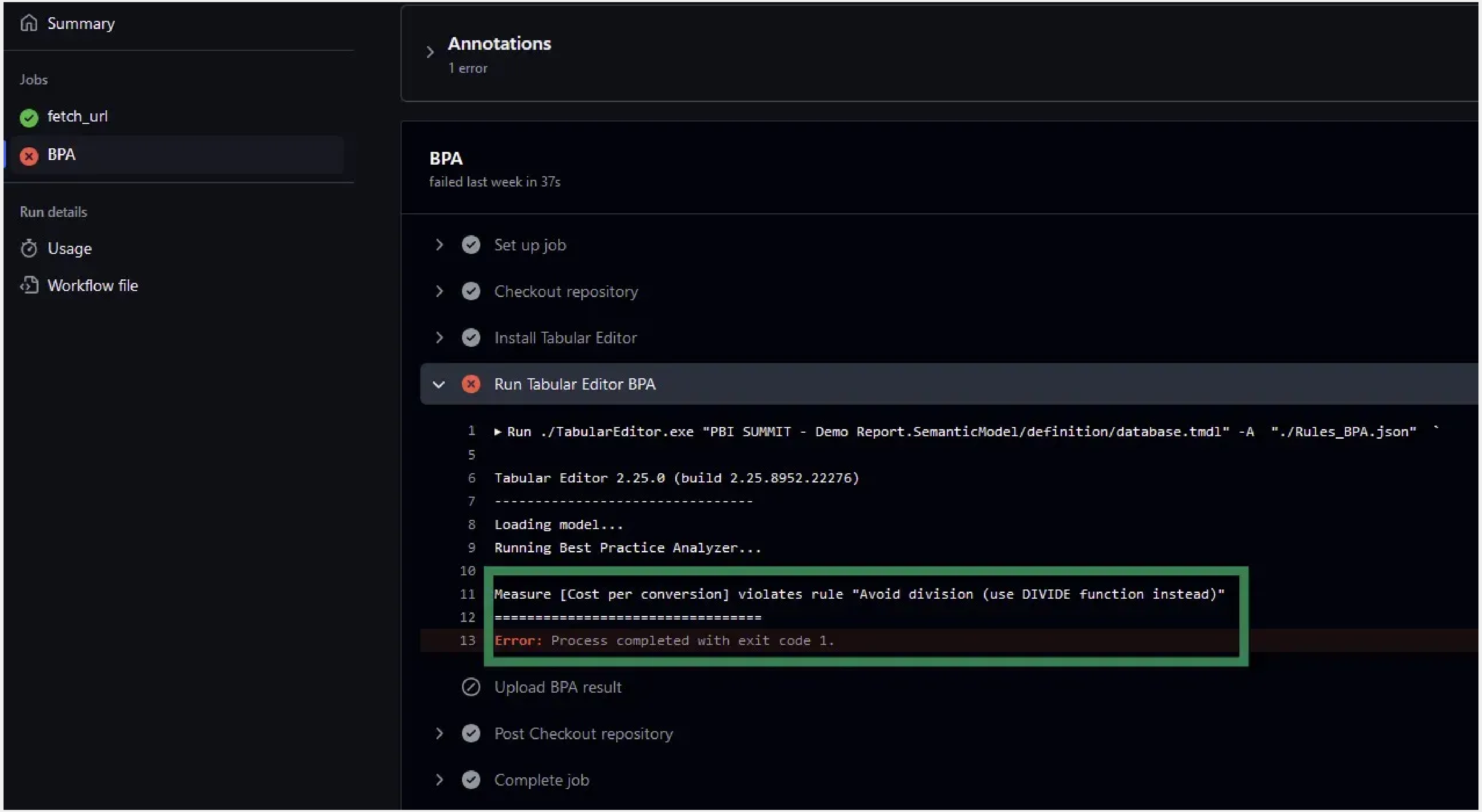
   *Only a handful of rules are tested and only one of the rules is an error (Severity 3)

Basic demo scenario

Link to demo video

🚧 Challange

❓ Run BPA checks automatically

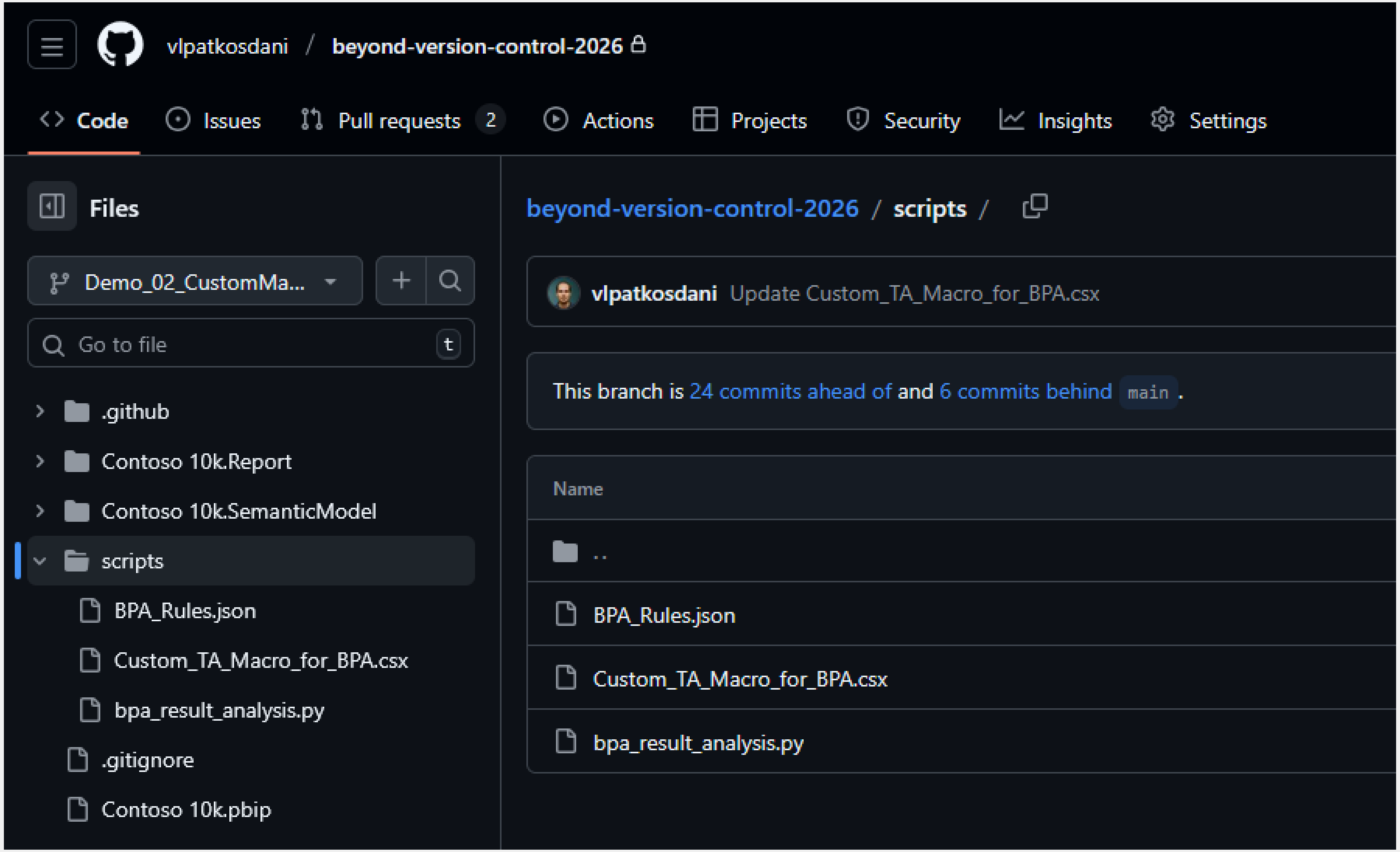

Summary
Jobs
fetch_url
BPA
Run details
Usage
Workflow file
Annotations
1 error
BPA
failed last week in 37s
Set up job
Checkout repository
Install Tabular Editor
Run Tabular Editor BPA
1 ▸ Run ./TabularEditor.exe "PBI SUMMIT - Demo Report.SemanticModel/definition/database.tmdl" -A "./Rules_BPA.json" `
5
6 Tabular Editor 2.25.0 (build 2.25.8952.22276)
7 --------------------------------
8 Loading model...
9 Running Best Practice Analyzer...
10
11 Measure [Cost per conversion] violates rule "Avoid division (use DIVIDE function instead)"
12 =================================
13 Error: Process completed with exit code 1.
Upload BPA result
Post Checkout repository
Complete job

🚧 Scaling Challange

❓ No clear indication what must be corrected

❓ Repeated runs for multiple issues

✅ Run BPA checks automatically

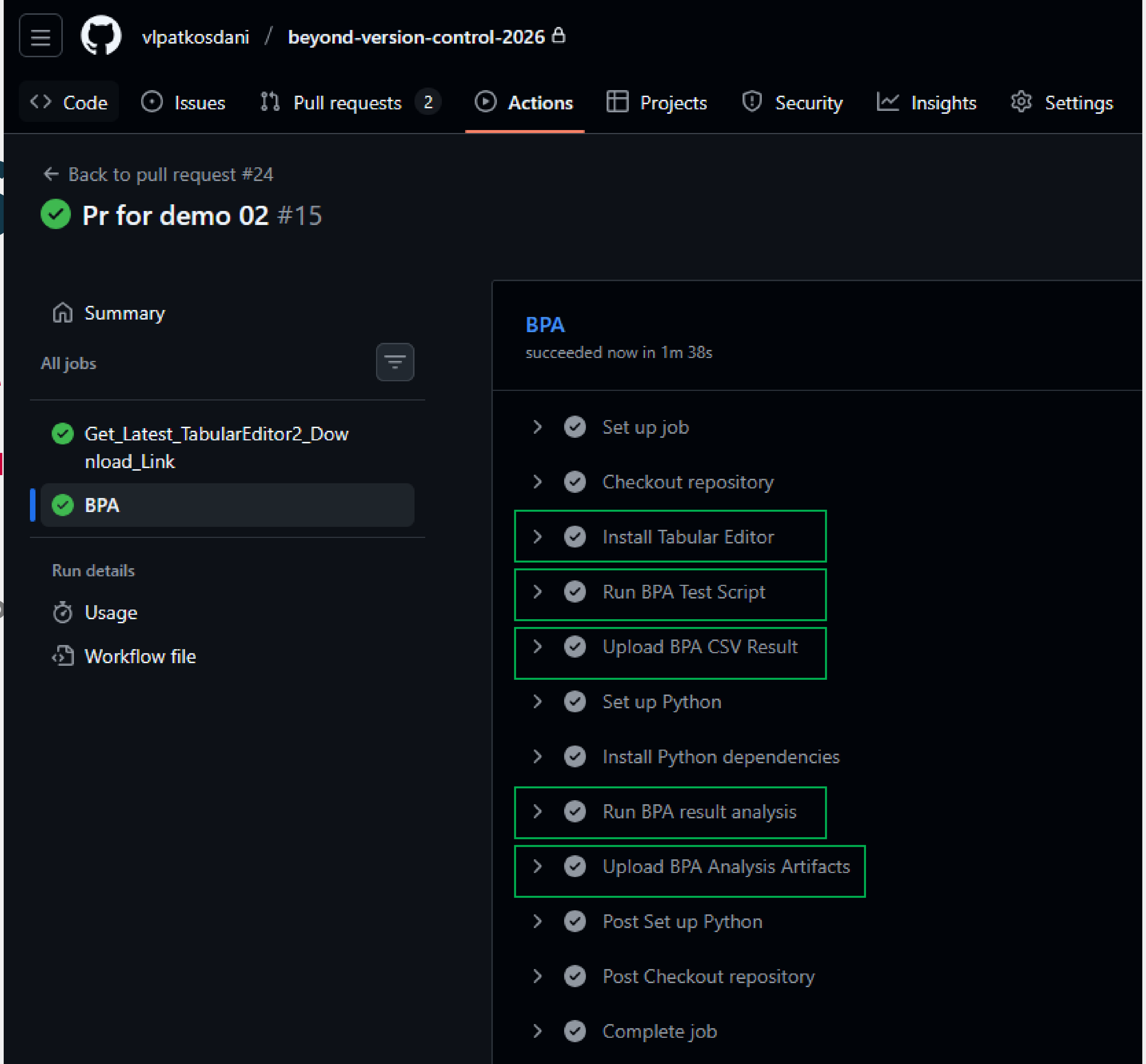

vlpatkosdani / beyond-version-control-2026
Code
Issues
Pull requests 2
Actions
Projects
Security
Insights
Settings
Files
Demo_02_CustomMa...
Go to file
t
.github
Contoso 10k.Report
Contoso 10k.SemanticModel
scripts
BPA_Rules.json
Custom_TA_Macro_for_BPA.csx
bpa_result_analysis.py
.gitignore
Contoso 10k.pbip
beyond-version-control-2026 / scripts /
vlpatkosdani Update Custom_TA_Macro_for_BPA.csx
This branch is 24 commits ahead of and 6 commits behind main .
Name
..
BPA_Rules.json
Custom_TA_Macro_for_BPA.csx
bpa_result_analysis.py

S

❓ No cle

❓ Repea

✅ Run BP



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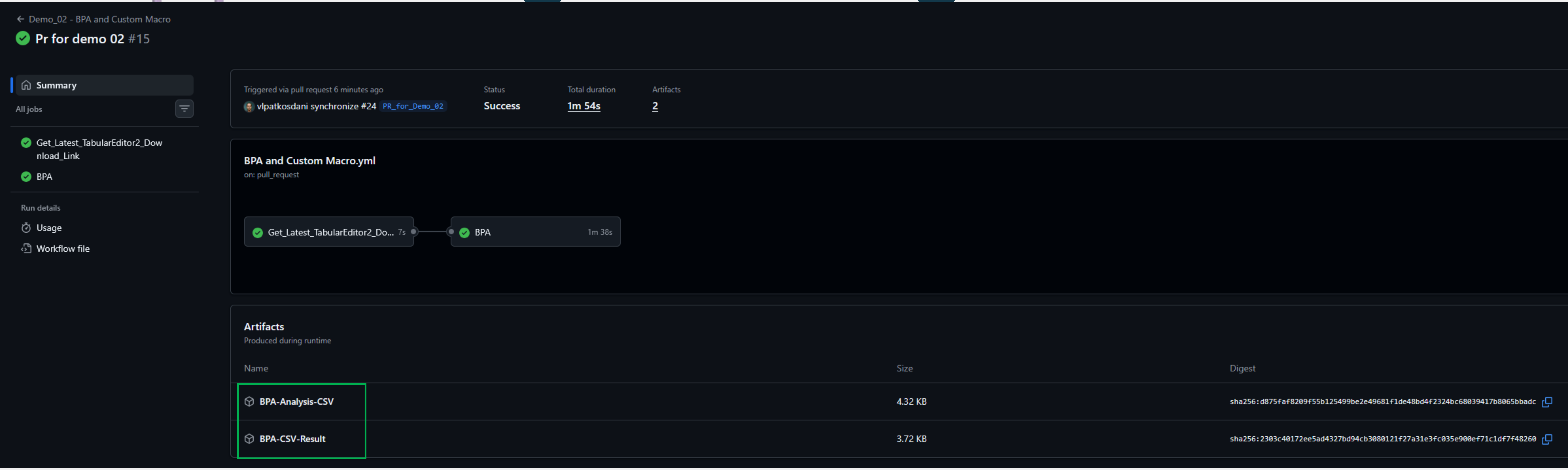
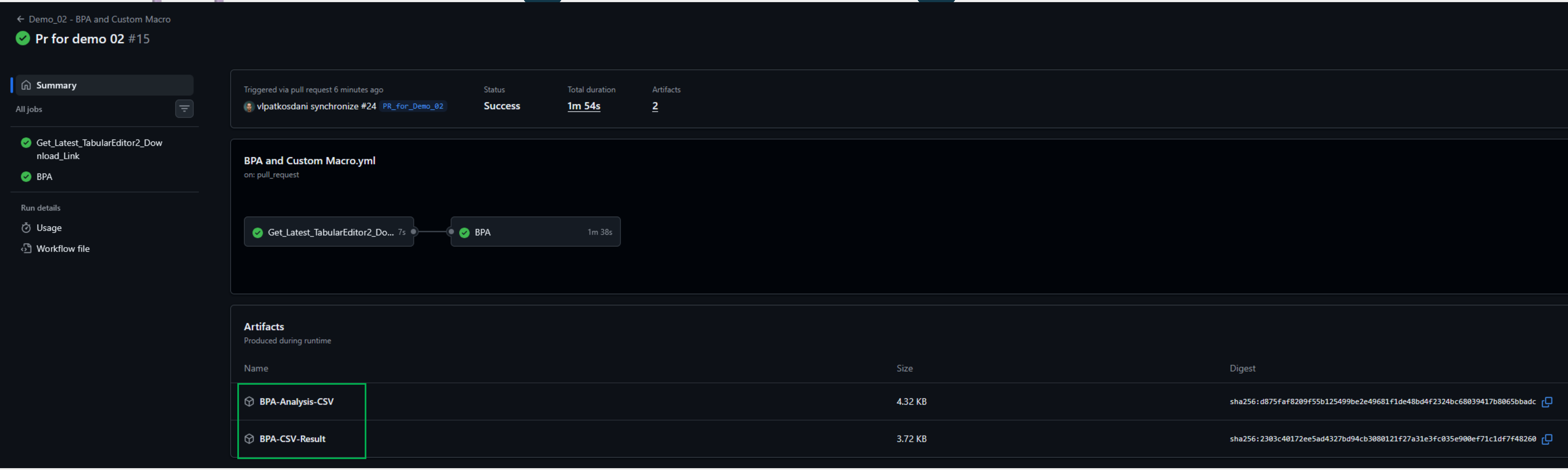
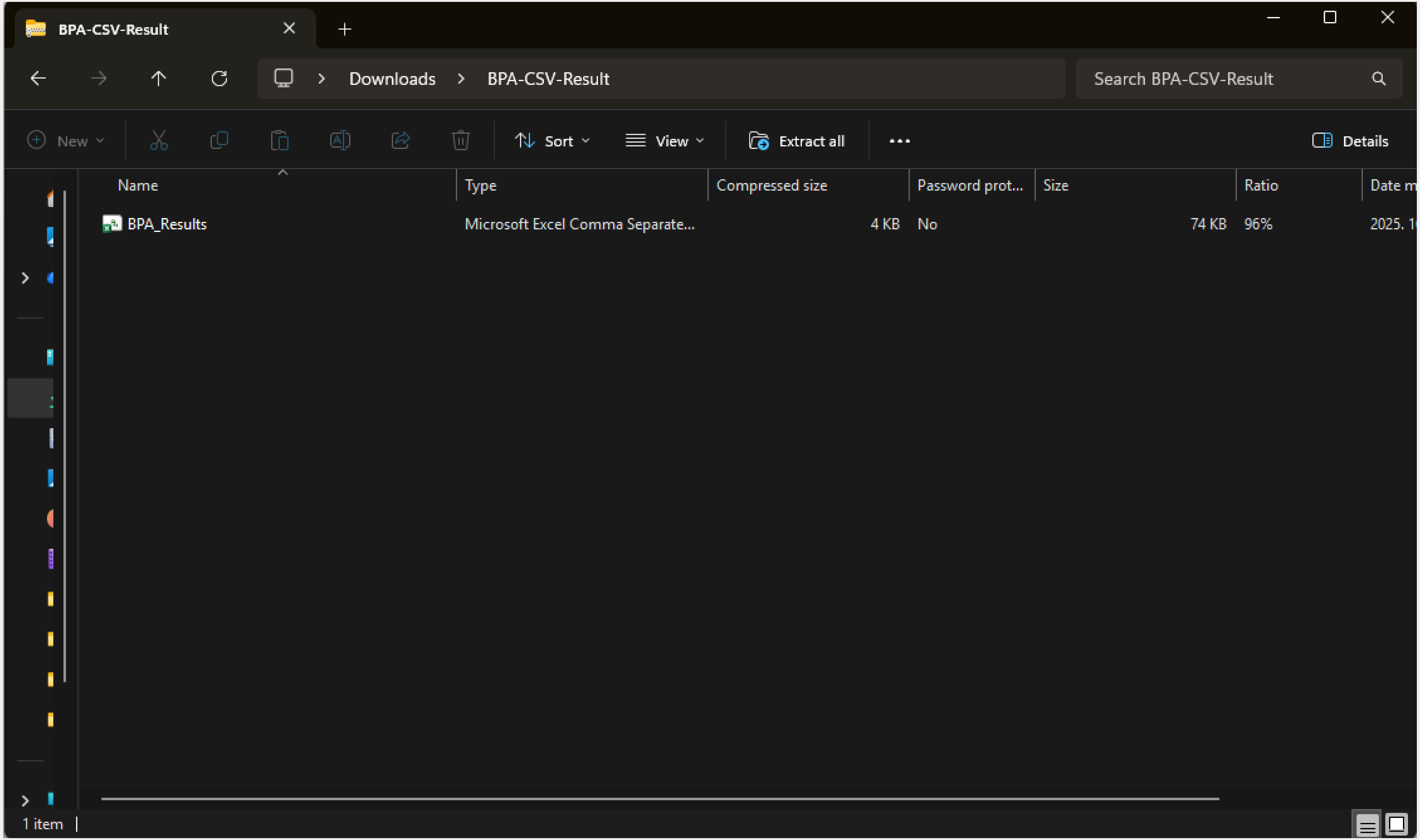


Scaling Challange




BPA-CSV-Result
Downloads
BPA-CSV-Result
Search BPA-CSV-Result
New
Sort
View
Extract all
Details
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Password prot...
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BPA_Results
Microsoft Excel Comma Separate...
4 KB
No
74 KB
96%
1 item

AutoSave Off BPA_Results - Read-Only

File | Home | Insert | Draw | Page Layout | Formulas | Data | Review | View | Automate | Add-ins | Help

Get & Transform Data: Get Data, From Text/CSV, From Web, From Table/Range, From Picture, Recent Sources, Existing Connections
Queries & Connections: Refresh All, Queries & Connections, Properties, Workbook Links
Data Types: Stocks (En..., Currencies..., Geography...
Sort & Filter: Sort, Filter, Clear, Reapply, Advanced
Data Tools: Text to Columns, Clean Data, Flash Fill, Remove Duplicates, Data Validation, Consolidate, Data Model, Analyze Data
Forecast: What-If Analysis, Forecast Sheet
Outline: Group, Ungroup, Subtotal

O18

	A	B	C	D	E	F	G	H
1	RuleCategory	RuleID	RuleName	RuleDescription	ObjectName	ObjectType	RuleSeverity	HasFixExpression
2	Formatting	5	[Formatting] Hide fact table columns	It is a best practice to hide fact table columns that are used for aggregation in measures.		Error	2	FALSE
3	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Customer'[CustomerKey]	Column	2	FALSE
4	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Customer'[Address]	Column	2	FALSE
5	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Customer'[State Code]	Column	2	FALSE
6	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Customer'[Zip Code]	Column	2	FALSE
7	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Customer'[Country Code]	Column	2	FALSE
8	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Customer'[Birthday]	Column	2	FALSE
9	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Date'[Working Day]	Column	2	FALSE
10	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Date'[Working Day Number]	Column	2	FALSE
11	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Product'[ProductKey]	Column	2	FALSE
12	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Product'[Weight Unit Measure]	Column	2	FALSE
13	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Product'[Weight]	Column	2	FALSE
14	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Product'[Unit Cost]	Column	2	FALSE
15	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Product'[Unit Price]	Column	2	FALSE
16	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[Order Number]	Column	2	FALSE
17	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[Line Number]	Column	2	FALSE
18	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[Order Date]	Column	2	FALSE
19	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[Delivery Date]	Column	2	FALSE
20	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[CustomerKey]	Column	2	FALSE
21	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[StoreKey]	Column	2	FALSE
22	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[ProductKey]	Column	2	FALSE
23	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[Quantity]	Column	2	FALSE
24	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[Unit Price]	Column	2	FALSE
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26	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[Unit Cost]	Column	2	FALSE
27	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[Currency Code]	Column	2	FALSE
28	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[Exchange Rate]	Column	2	FALSE
29	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Store'[StoreKey]	Column	2	FALSE
30	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Store'[Store Code]	Column	2	FALSE
31	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Store'[Open Date]	Column	2	FALSE
32	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Store'[Close Date]	Column	2	FALSE
33	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Currency Exchange'[Date]	Column	2	FALSE
34	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Currency Exchange'[FromCurrency]	Column	2	FALSE
35	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Currency Exchange'[ToCurrency]	Column	2	FALSE
36	Metadata	7	Disable attribute hierachies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Currency Exchange'[Exchange]	Column	2	FALSE
37	Performance	8	[Performance] Set IsAvailableInMdx to false on non-attribute columns	To speed up processing time and conserve memory after processing, attribute hierarchies	'Customer'[CustomerKey]	Column	2	FALSE
38	Performance	8	[Performance] Set IsAvailableInMdx to false on non-attribute columns	To speed up processing time and conserve memory after processing, attribute hierarchies	'Customer'[Address]	Column	2	FALSE
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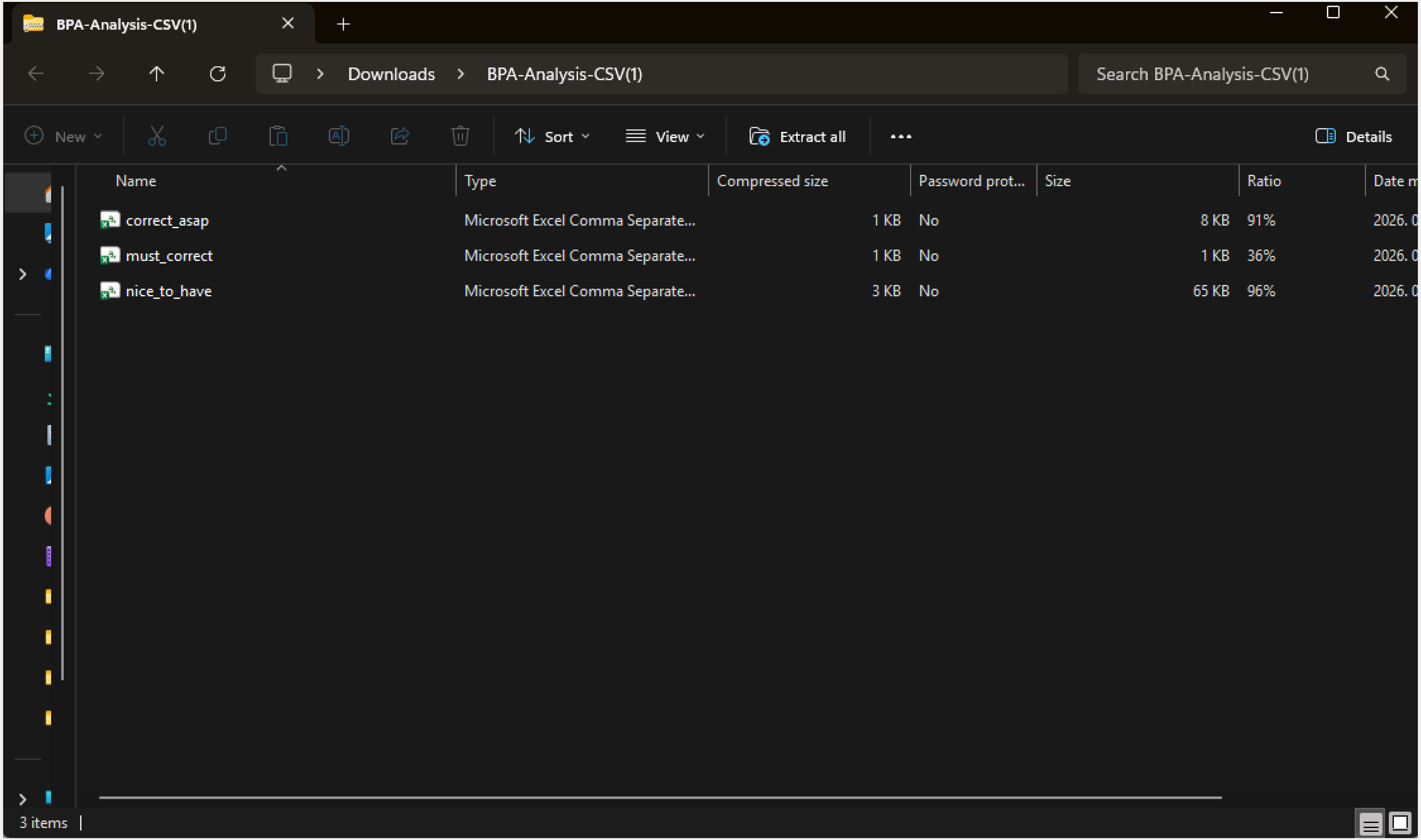
43	Performance	8	[Performance] Set IsAvailableInMdx to false on non-attribute columns	To speed up processing time and conserve memory after processing, attribute hierarchies	'Date'[Working Day]	Column	2	FALSE
44	Performance	8	[Performance] Set IsAvailableInMdx to false on non-attribute columns	To speed up processing time and conserve memory after processing, attribute hierarchies	'Date'[Working Day Number]	Column	2	FALSE
45	Performance	8	[Performance] Set IsAvailableInMdx to false on non-attribute columns	To speed up processing time and conserve memory after processing, attribute hierarchies	'Product'[ProductKey]	Column	2	FALSE
46	Performance	8	[Performance] Set IsAvailableInMdx to false on non-attribute columns	To speed up processing time and conserve memory after processing, attribute hierarchies	'Product'[Weight Unit Measure]	Column	2	FALSE

BPA_Results

Ready | Accessibility: Unavailable | Display Settings | 100%


BPA-Analysis-CSV(1)
Downloads > BPA-Analysis-CSV(1)
Search BPA-Analysis-CSV(1)
New
Sort
View
Extract all
Details
Name
Type
Compressed size
Password prot...
Size
Ratio
Date m
correct_asap
Microsoft Excel Comma Separate...
1 KB
No
8 KB
91%
2026. 0
must_correct
Microsoft Excel Comma Separate...
1 KB
No
1 KB
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2026. 0
nice_to_have
Microsoft Excel Comma Separate...
3 KB
No
65 KB
96%
2026. 0
3 items

must_correct - Read-Only

RuleCategory	RuleID	RuleName	ObjectName	ObjectType	Custom_Severity	Description
DAX Expressions	52	Avoid division (use DIVIDE function instead)	[Rule Violation - Division]	Measure	3	Calculated Columns, Measures or Calculated Tables should not use the division symbol in their expressions (/) unless the denominator is a constant value. Instead, it is advised to always use the DIVIDE(<numerator>,<denominator>) function.

must_correct

Demo

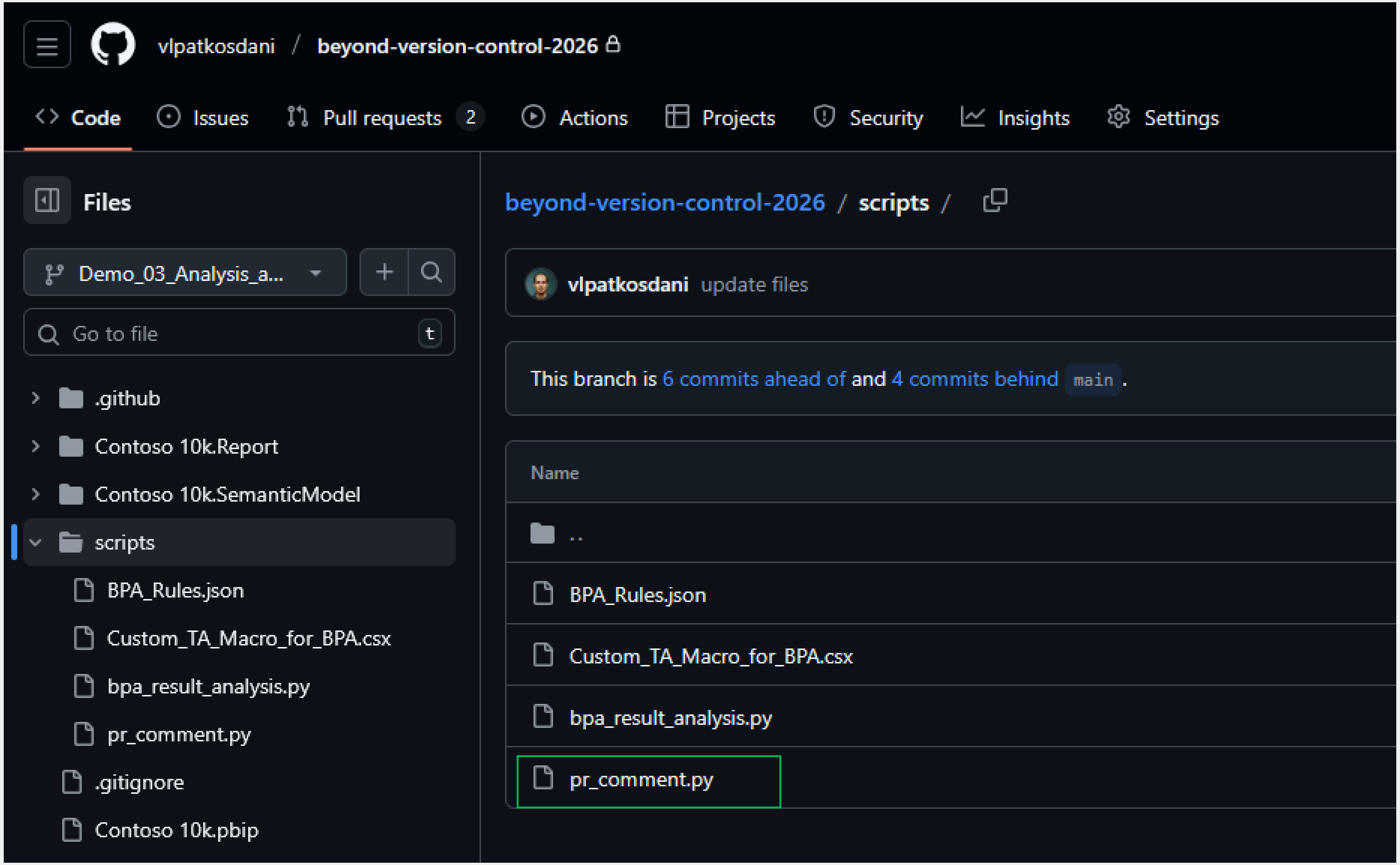
- ✅ No clear indication what must be corrected
- ✅ Repeated runs for multiple issues

- ✅ Run BPA checks automatically

🚧 Scaling challenges

❓ Focus on important issues first
❓ Get additional context
❓ Lack of detailed feedback on Pull Request

✅ No clear indication what must be corrected
✅ Repeated runs for multiple issues
✅ Run BPA checks automatically



vlpatkosdani / beyond-version-control-2026
Code
Issues
Pull requests 2
Actions
Projects
Security
Insights
Settings
Files
Demo_03_Analysis_a...
Go to file
t
.github
Contoso 10k.Report
Contoso 10k.SemanticModel
scripts
BPA_Rules.json
Custom_TA_Macro_for_BPA.csx
bpa_result_analysis.py
pr_comment.py
.gitignore
Contoso 10k.pbip
beyond-version-control-2026 / scripts /
vlpatkosdani update files
This branch is 6 commits ahead of and 4 commits behind main .
Name
..
BPA_Rules.json
Custom_TA_Macro_for_BPA.csx
bpa_result_analysis.py
pr_comment.py

github-actions bot commented 2 minutes ago

❌ Model cannot be deployed until Severity 3 issues are fixed.

📁 **BPA Result & Result analysis (CSV downloads):** [View .zip files Here]

📊 Summary

Severity Level	Number of Issues
🚨 Must Correct	1
⚡ Correct ASAP	41
💡 Nice to Have	201

🚨 Must Correct (Severity 3)

RuleName	Violations
Avoid division (use DIVIDE function instead)	1

▾ 🔎 **Avoid division (use DIVIDE function instead) – Rule Description** 🔗

- **[Rule Violation - Division]** (Measure)

⚡ Correct ASAP (Severity 2)

RuleName	Violations
Disable attribute hierachies to decrease processing	34

Demo

- ✅ Lack of detailed feedback on Pull Request
- ✅ Focus on important issues first
- ✅ Give additional context

- ✅ No clear indication what must be corrected
- ✅ Repeated runs for multiple issues
- ✅ Run BPA checks automatically

🌳 Scalable CI/CD workflows

Answering multi-project environment’s needs

🚧 Scaling challenges

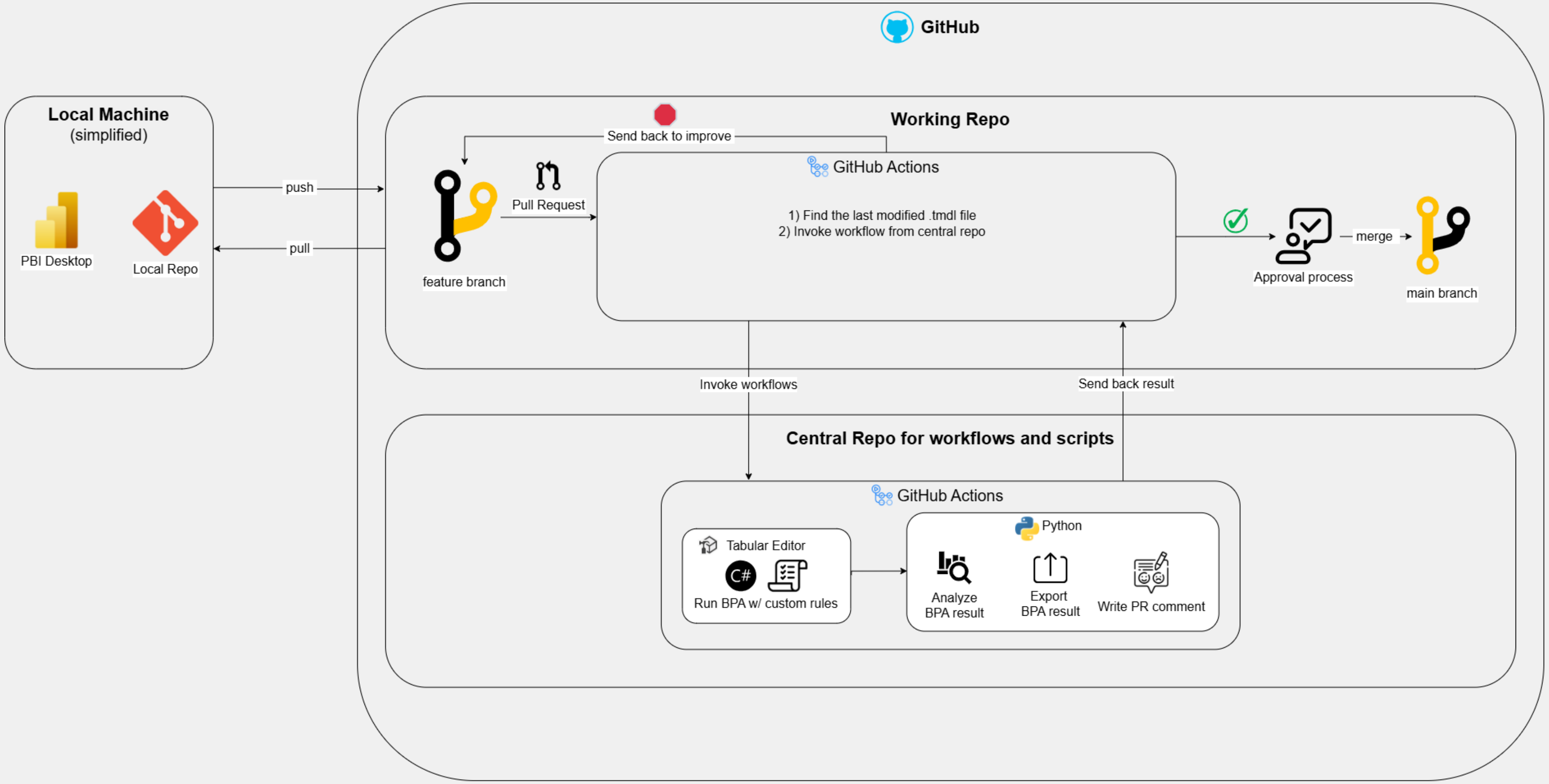
- ❓ Version control of the solution package
- ❓ Automatically create GA workflow to each newly created repository
- ❓ Set default approver(s)
- ❓ Control which part(s) of the workflow you need to run based on PR

Reusable workflow across the organization

✅ Version control of the solution package

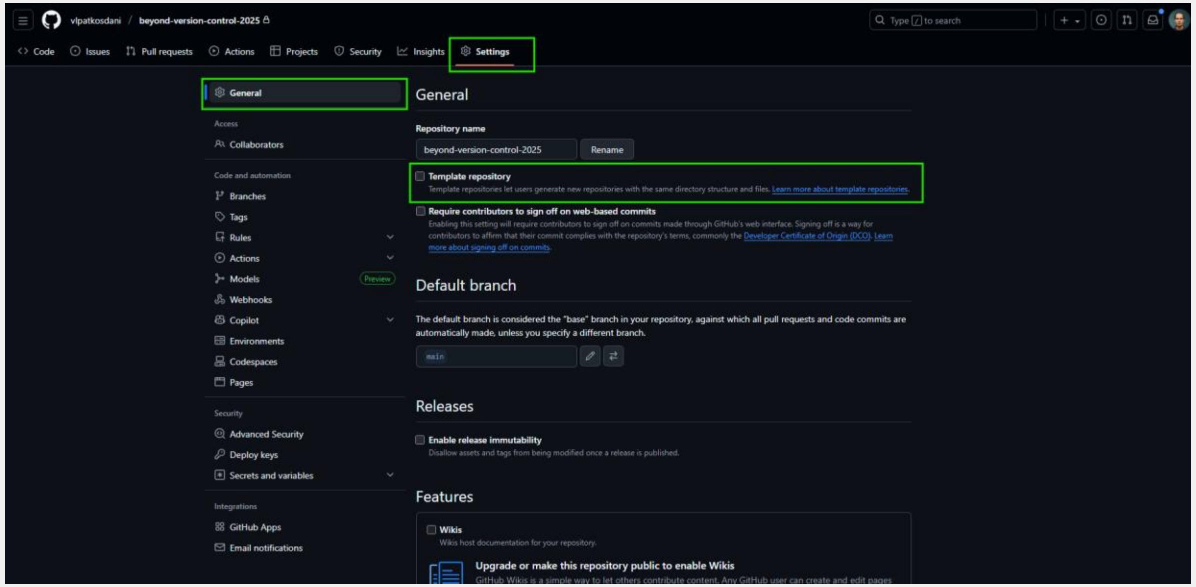
```
run-bpa:
  needs: find-latest-tmdl
  uses: Your-Organization/central-repo-for-automated-pr-checks/.github/workflows/Reusable_BPA_Checker.yml@main
  with:
    tmdl_path: ${{ needs.find-latest-tmdl.outputs.tmdl_path }}
  secrets:
    Central_Repo_PAT: ${{ secrets.Central_Repo_PAT }}
```

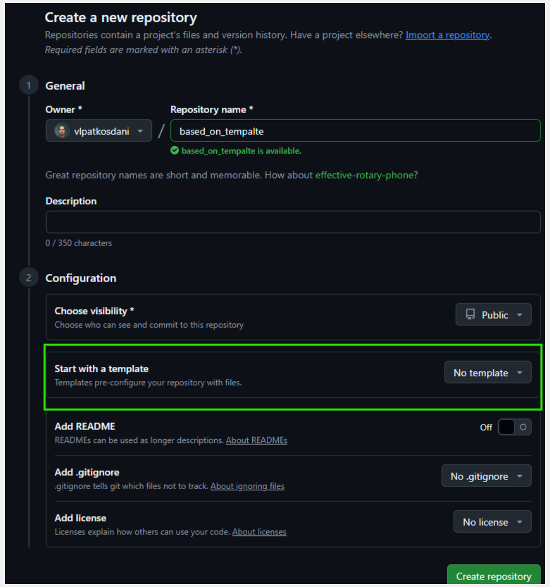
PR triggered BPA – stored in central repo



Template repo

✅ Add workflow for each newly created repository





Assign default approvers 1

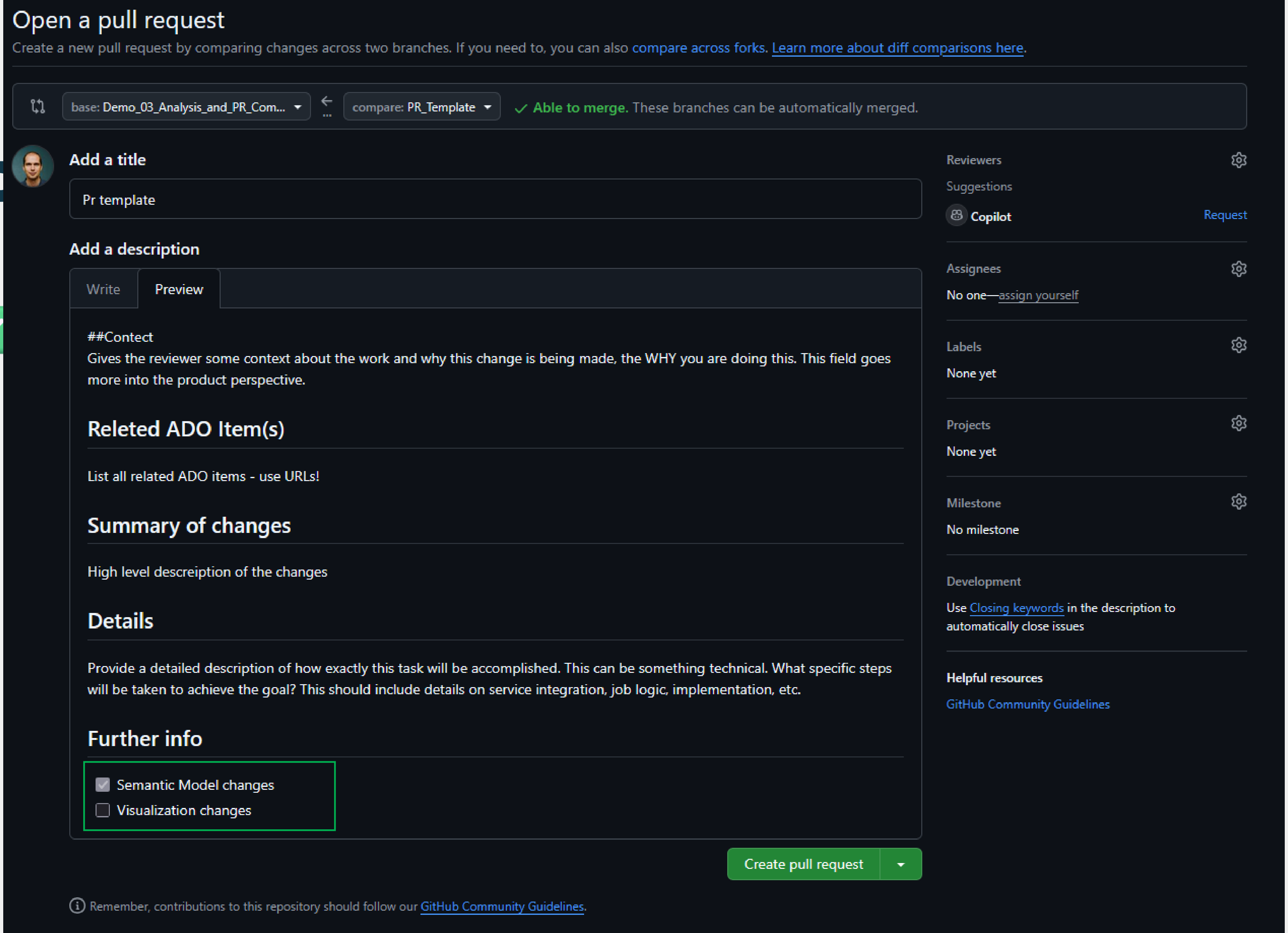
✅ Set default approvers

```
assign_reviewer:
  needs: run-bpa
  if: github.event_name == 'pull_request'
  runs-on: ubuntu-latest
  steps:
    - name: Assign Reviewer
      run: |
        curl -X POST \
          -H "Authorization: token ${{ secrets.GITHUB_TOKEN }}" \
          -H "Accept: application/vnd.github.v3+json" \
          https://api.github.com/repos/${{ github.repository }}/pulls/${{ github.event.pull_request.number }}/requested_reviewers \
          -d '{"reviewers": ["vlpatkosdani"]}'
```

Assign default approvers 2

✅ Set default approvers

Add CODEOWNERS file in the repo + set up branch protection rules

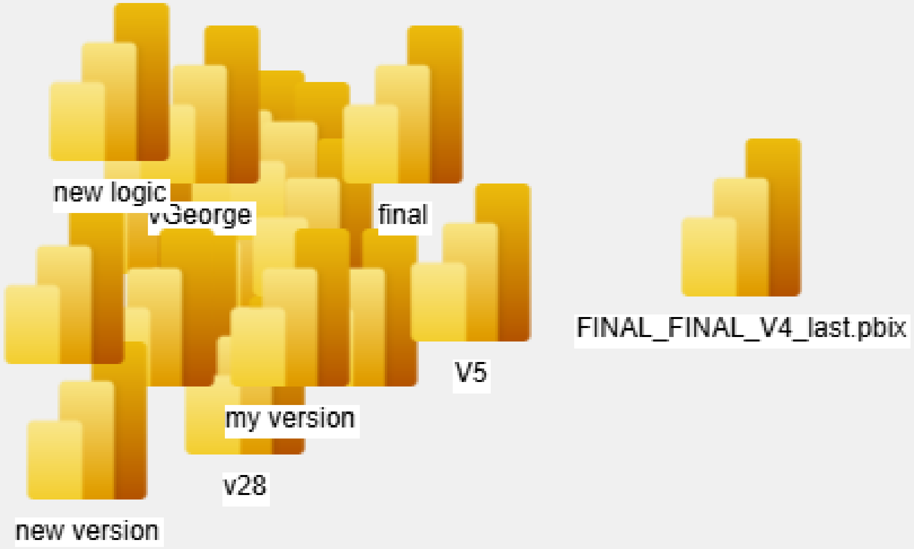

Open a pull request
Create a new pull request by comparing changes across two branches. If you need to, you can also compare across forks. Learn more about diff comparisons here.
base: Demo_03_Analysis_and_PR_Com...
compare: PR_Template
Able to merge. These branches can be automatically merged.
Add a title
Pr template
Add a description
Write
Preview
##Contect
Gives the reviewer some context about the work and why this change is being made, the WHY you are doing this. This field goes more into the product perspective.
Releted ADO Item(s)
List all related ADO items - use URLs!
Summary of changes
High level descreiption of the changes
Details
Provide a detailed description of how exactly this task will be accomplished. This can be something technical. What specific steps will be taken to achieve the goal? This should include details on service integration, job logic, implementation, etc.
Further info
Semantic Model changes
Visualization changes
Create pull request
Remember, contributions to this repository should follow our GitHub Community Guidelines.
Reviewers
Suggestions
Copilot
Request
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone
No milestone
Development
Use Closing keywords in the description to automatically close issues
Helpful resources
GitHub Community Guidelines

Monitor all repos

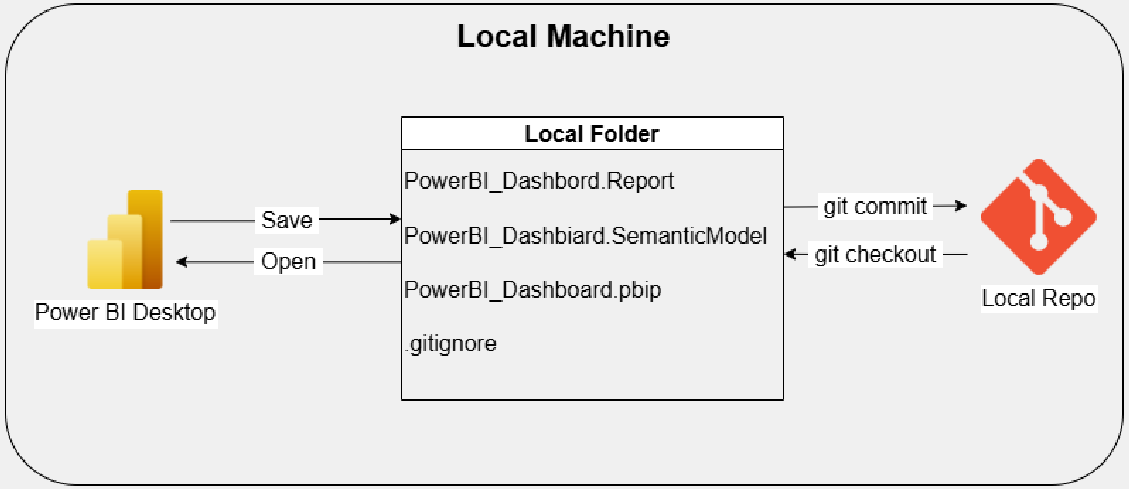
- Send logs of all runs into a central repo
- Understand the state, strength and areas of improvements of your models across the entire organization and your team

Wrap up

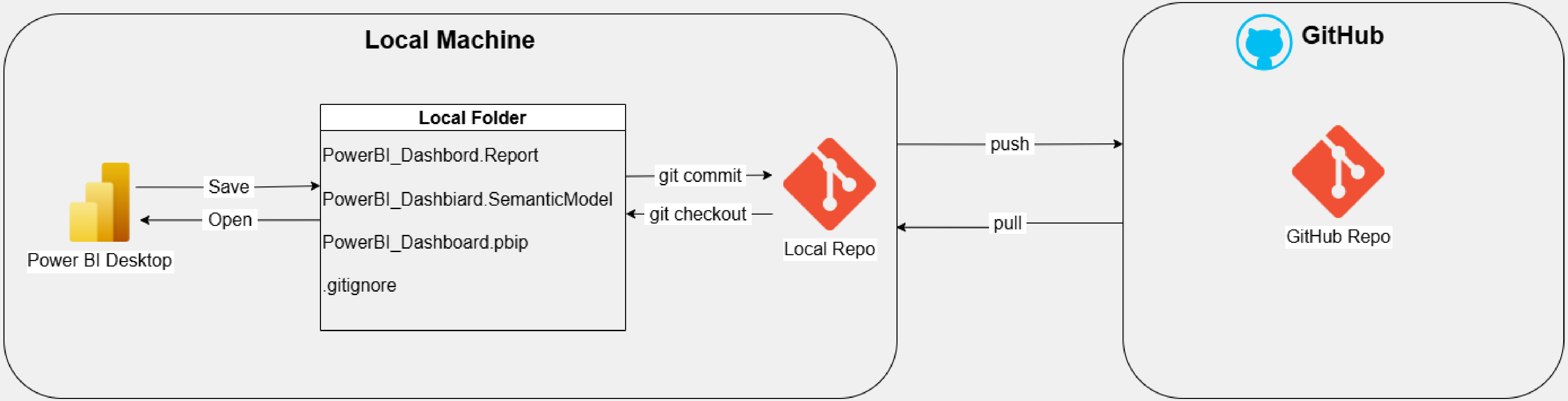
Version Chaos



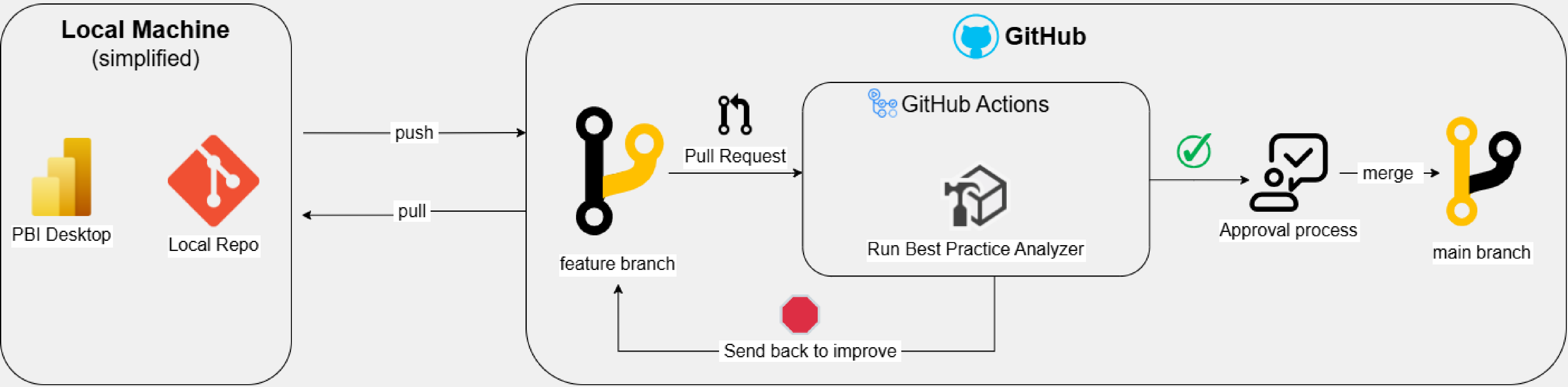
Local version control



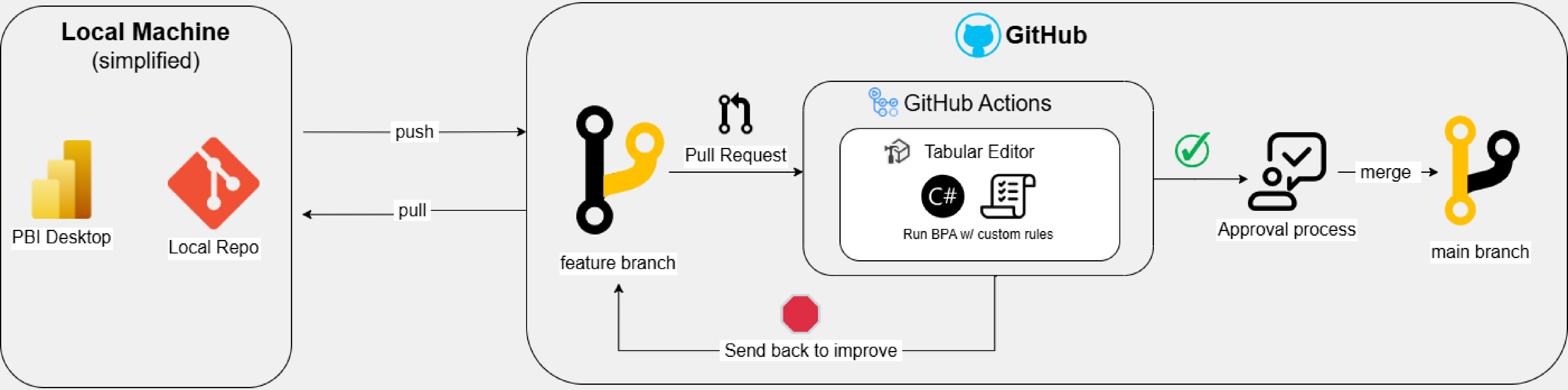
Repo stored on GitHub



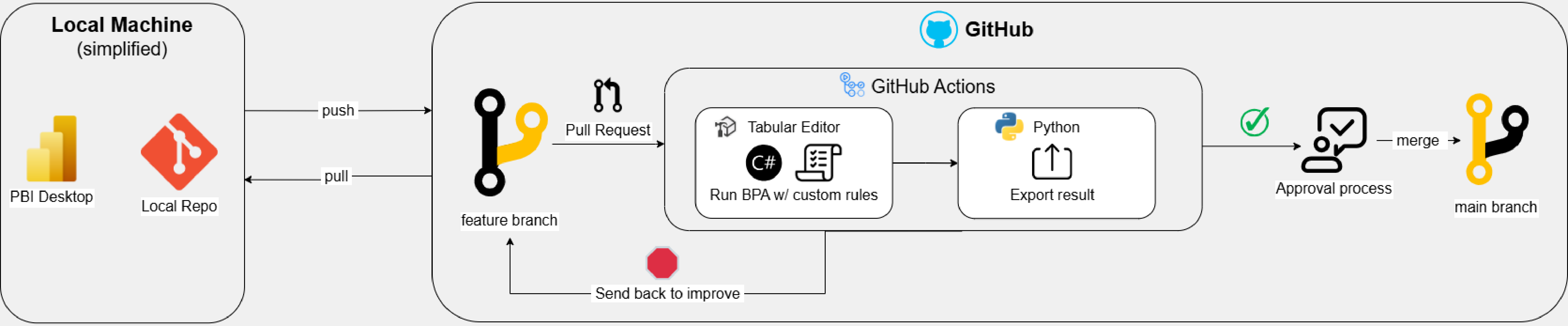
PR triggered BPA analysis



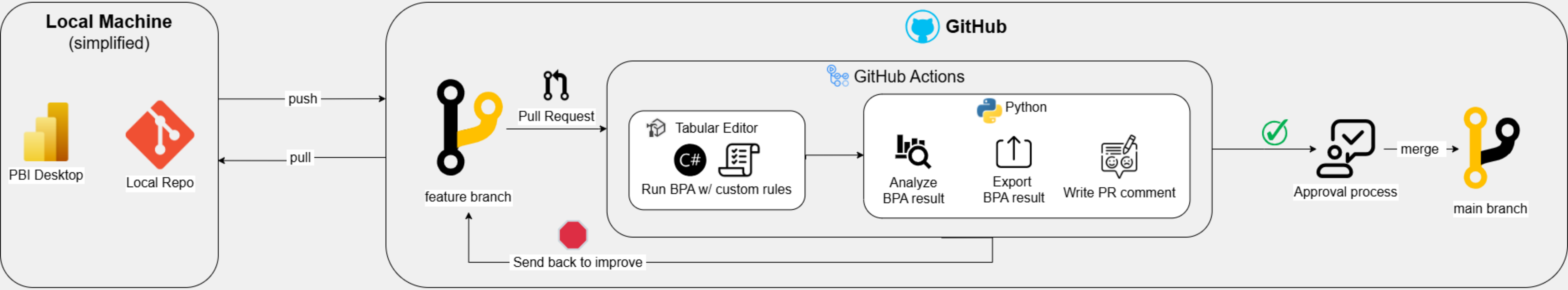
PR triggered BPA – custom rules



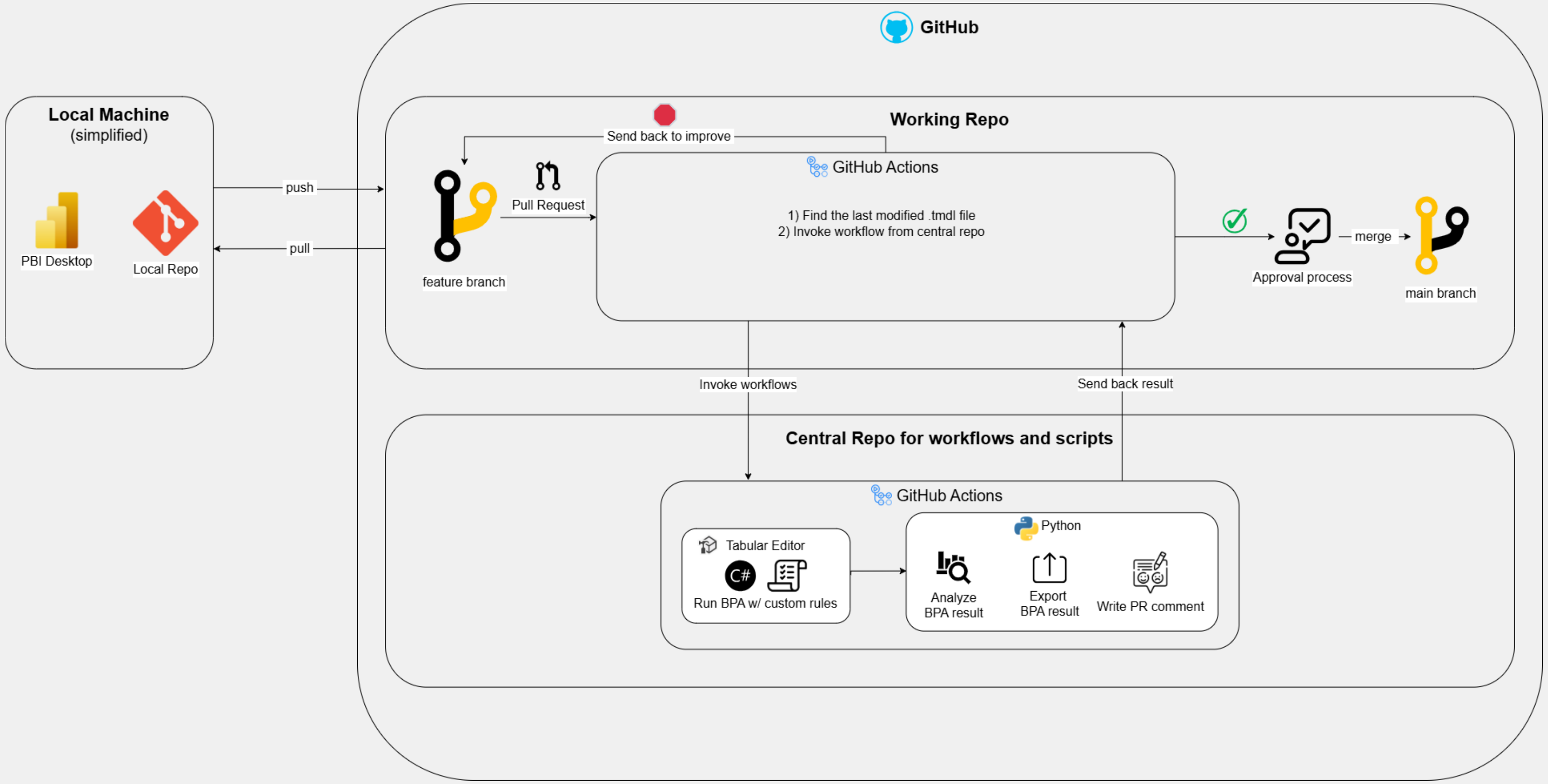
PR triggered BPA – export results



PR triggered BPA – detailed PR message



PR triggered BPA – stored in central repo



Reminder

Make sure everyone is using the way you intended and everyone is are comfortable using it

Start using it now!

Resources to study more

[Kevin Chant]

[Peer Grønnerup]

[Rui Romano]

[Mathias Thierbach]

Check exmaple repo and realted articles

Building your Power BI CI/CD pipeline with GitHub Actions (Part 1)

Automating BPA with C# in Tabular Editor + GitHub Actions (Part 2)

Making BPA Results Actionable with Automated PR Comments (Part 3)

Scaling Your Power BI CI/CD Pipeline Across Projects (Part 4)

GitHub Repo

Time for: thoughts, questions, or reactions